

Project Title:
E-Commerce Product Entity-Resolution & Matching Platform
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

E-commerce platforms constantly face the challenge of identifying when listings from different sellers, marketplaces, or catalogs refer to the same underlying product, despite inconsistent titles, attributes, images, and missing identifiers. This platform addresses that problem through a hybrid entity-resolution pipeline that combines rule-based matching (SKUs, UPCs/GTINs, brand-model patterns) with fuzzy string matching, semantic embeddings, and image-based visual similarity.

A blocking and candidate-generation stage narrows the comparison space before pairwise or clustering-based matching, allowing the system to scale efficiently to catalogs containing millions of SKUs. The resulting outputs include unified product IDs, deduplicated catalogs, and confidence-scored match links, which support downstream applications such as price comparison, competitive intelligence, and marketplace onboarding.

The architecture integrates data ingestion, feature extraction, a scoring engine, and a human-in-the-loop review interface for low-confidence matches. Key design considerations include multilingual listings, variant differentiation, and balancing precision against recall to minimize false merges and missed matches.

Signature of the Guide